INCH-POUND

MIL-R-39023A
AMENDMENT 2
25 July 1994
SUPERSEDING
AMENDMENT 1
13 April 1979

MILITARY SPECIFICATION

RESISTORS, VARIABLE, NON-WIRE-WOUND, PRECISION
GENERAL SPECIFICATION FOR

This amendment forms a part of MIL-R-39023A, dated 3 June 1974, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 1

1.2.1 Type designation. Under first "A", delete and substitute: "Resistance temperature characteristic, maximum ambient temperature and taps (1.2.1.2)."

PAGE 2

1.2.1.2, delete and substitute:

"1.2.1.2 Resistance temperature characteristic, maximum ambient temperature and taps. The resistance temperature characteristic, maximum ambient temperature and tap is identified by a single letter in accordance with tables I and IA."

TABLE I, delete substitute:

"TABLE I. Resistance temperature characteristic, maximum abient temperature and taps. 1/

Symbol	Ambient temperature with rated load in °C	Ambient temperature with zero load in °C	Tap located at center of resistance element (±1°C) 2/
A	70	125	Not applicable
B	70	125	Applicable

1/ The test temperature and maximum resistance change shall be as specified in table IA.

2/ Applicable only to linear function resistors."

PAGE 5

*2.1, SPECIFICATIONS, MILITARY: Delete "MIL-C-45662" and corresponding title.

*2.1, STANDARDS, MILITARY: Delete "MIL-STD-105" and corresponding title.

*2.1, STANDARDS, MILITARY: Add "MIL-STD-45662 - Calibration Systems Requirements."

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3.14, Taps, line 4, delete "(see 4.3.2, 3.1, and 6.1.2)" and substitute "(see 1.2.1.2, table I, 3.1, 4.3.2, and 6.1.2)".

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*4.1.1, last sentence, delete "MIL-C-45662" and substitute "MIL-STD-45662".

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4.4.4, delete and substitute:

"4.4.4 Retention of qualification. Every 6 months, the manufacturer shall compile a summary of the results of quality conformance inspection in the form of a retention of qualification report, and forward it to the qualifying activity within 30 days from the end of the reporting period as the basis of continued qualification approval. In addition, the manufacturer shall immediately notify the qualifying activity whenever the group B inspection data indicates failure of the qualified product to meet the requirements of this specification. Continuation shall be based on the evidence that over the 6-month period the following has been met:

a. The manufacturer has not modified the design of the item.

b. The specification requirements for the item have not been amended so far as to affect the character of the item.

c. Lot rejection for group A inspection does not exceed the AQL of table VIII.

d. The requirements for group B inspection are met.

When group B requirements were not met and the manufacturer has taken corrective action satisfactory to the government, group B retesting shall be instituted. A summary of the retesting shall be forwarded to the qualifying activity within 30 days after completion of the retest. All reports are to be certified by a responsible company official and the government inspector."

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*TABLE X, delete and substitute:

"TABLE X. Group A inspection.

Test	Requirement paragraph	Method paragraph	Number of samples
Dimensions	3.1	4.6.1.1	
Lateral runout	3.7	4.6.3	
Shaft runout	3.8	4.6.4	
Pilot-diameter runout	3.9	4.6.5	
Shaft radial play	3.10	4.6.6	
Shaft end play	3.11	4.6.7	
Torque	3.13.1	4.6.9.1	
Marking	3.33	4.6.1.1	
Workmanship	3.34 to 3.34.2 incl.	4.6.1.1	See
DC resistance	3.6	4.6.2	4.5.1.2.1
Taps (when applicable)	3.14	4.6.10	
Dielectric withstanding voltage (atmospheric)	3.15	4.6.11	
Insulation resistance	3.16.1	4.6.12	
Function conformity tolerance	3.17.1.1	4.6.13	
Mechanical backlash	3.18	4.6.14	
End voltage	3.20	4.6.16	
Output smoothness	3.21.1	4.6.17	
Terminal strength	3.22	4.6.18	

"

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*Following TABLE X, add the following new table:

"TABLE X-1. Group A sampling plan.

Lot size	Sample size
2 to 13	100%
14 to 150	13
151 to 280	20
281 to 500	29
501 to 1200	34
1201 to 3200	42
3201 to 10000	50
10001 to 35000	60
35001 to 150000	74
150001 to 500000	90
500001 and over	102

"

*4.5.1.2.1, delete and substitute:

"4.5.1.2.1 Sampling plan. A sample of parts shall be randomly selected in accordance with table X-1, if one or more defects are found, the lot shall be rescreened and defects removed. After screening and removal of defects, a new sample of parts shall be randomly selected in accordance with table X-1, if one or more defects are found in the second sample, the lot shall be rejected and shall not be supplied to this specification."

*4.5.1.2.2, delete in its entirety.

The margins of this amendment are marked with an asterisk to indicate where changes from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - AR, AT, AV, ME, MI
Navy - AS, CG, MC, OS
Air Force - 17, 19, 80
DLA - ES

Preparing activity:
Army - ER

Agent:
DLA - ES

(Project 5905-1379)

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